

Sumeet Parameshwar Shirgure

PHD STUDENT · QUANTUM COMPUTING

University of Central Florida

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Education

University of Central Florida

PHD IN COMPUTER ENGINEERING

Orlando, FL

August 2025 - Present

- Advisor: Dr. Siyuan Niu
- Research focus : Quantum Computing and Quantum Information, Computer Architecture

University of Southern California

MS IN COMPUTER SCIENCE

Los Angeles, CA

September 2021 - May 2023

- Supervisor: Dr. Massoud Pedram
- Project description : VLSI EDA tools for Superconductor Electronic Logic

Indian Institute of Technology

B.TECH(HONS.) + M.TECH IN COMPUTER SCIENCE AND ENGINEERING

Kharagpur, WB

July 2015 - April 2020

- Graduated with honors, and a cumulative GPA of 9.06/10.00

Professional Experience

Ming Hsieh Department of Electrical and Computer Engineering, USC

RESEARCH ASSISTANT

Los Angeles, CA

May 2022 - April 2023

- Development of EDA tools for superconductor electronics

Google

SOFTWARE ENGINEER

Bengaluru, India

August 2020 - June 2021

- Software Engineer in the call ads infrastructure team

D.E Shaw

SOFTWARE ENGINEERING INTERN

Hyderabad, India

May 2019 - July 2019

- Built machine learning infrastructure

Adobe

DEEP LEARNING RESEARCH INTERN

Noida, India

May 2018 - July 2018

- Research on neural language models using recurrent neural networks

Publications

PUBLISHED

Sumeet Shirgure and Siyuan Niu. Error Mitigation in Dynamic Circuits for Hamiltonian Simulation. In Proceedings of the Great Lakes Symposium on VLSI 2026 (GLSVLSI '26). Association for Computing Machinery, New York, NY, USA, 843–849

Sumeet Shirgure, Efehan Kökcü, Anupam Mitra, Wibe Albert de Jong, Costin Iancu, Siyuan Niu. Scalable Benchmarking Framework for Dynamic Quantum Circuits. To appear in 59th IEEE/ACM International Symposium on Microarchitecture.

IN REVIEW / IN PREPARATION

Sumeet Shirgure, Efehan Kökcü, Siyuan Niu. Challenges in Barren Plateau Mitigation with Dynamic Parameterized Quantum Circuits. arXiv preprint arXiv:2606.23751 (2026). With the editors of *Quantum*.

Sumeet Shirgure, Yanzhang Zhu, Di Wu, Siyuan Niu. Speculative Matching for Low Latency Surface Code Quantum Error Correction.

Awards, Fellowships, & Grants

2015	Ranked 415th among 1.5 million applicants , IIT Joint Entrance Exam (Advanced)	India
2019	Top 8 teams from the Indian subcontinent , International Collegiate Programming Contest	Porto
2019	Top 1000 participants worldwide , Google's CodeJam	Online
2019	Top 500 participants worldwide , Meta's HackerCup	Online
2020	Scored 325/340 , GRE, Verbal: 157/170, Quantitative: 168/170, 4.0/6.0 Analytical writing	India
2021	Dean's Master's scholarship , \$5000 fee waiver	USC
2022	Quantum computing hackathon winner , QCHack	Online
2022	Accepted into the CS MS honors program ,	USC
2022	Graduate RA award , Full tuition fee waiver and stipend	USC
2023	Initiated into Phi Kappa Phi , Top 10% of graduate students	USC
2023	MIT iQuHACK participant , Invited to the in-person quantum computing hackathon.	MIT
2025	ORCGS Doctoral Fellowship , Full tuition fee waiver and stipend	UCF

Presentations

CONTRIBUTED TALKS

Scalable Benchmarking Framework for Dynamic Quantum Circuits
59th IEEE/ACM International Symposium on Microarchitecture, November 2026, Athens, Greece

Challenges in Barren Plateau Mitigation with Dynamic Parameterized Quantum Circuits
Quantum Techniques in Machine Learning, December 2026, Stellenbosch University, South Africa

POSTER PRESENTATIONS

A flaw in the NeurIPS unlearning challenge and an algorithmic framework for entropy regularization
Neural Information Processing Systems, December 2023, New Orleans, LA

Dynamic Circuit Compilation for Sparse Qubit Connectivity
IEEE International Conference on Quantum Computing and Engineering, September 2025, Albuquerque, USA
Adaptive Quantum Circuits, November 2025, Boston MA, USA

Teaching Experience

Fall 2019	Advanced Graph Theory , Teaching Assistant	IIT Kharagpur
Fall 2025	Quantum Computing Systems , Teaching Assistant	UCF

Extracurricular Activities

SERVICE AND OUTREACH

2015-2017	National Service Scheme , Student Volunteer	IIT Kharagpur
2026	Qiskit Fall Fest 2026 at UCF , Organizer	UCF

PEER REVIEW

Sub-reviewer for GLS VLSI 2026